

PYTHON DAY 05

STRING OPERATIONS • 90 DAYS CHALLENGE

01 What is a String?

A string is a sequence of characters enclosed in quotes. Strings are one of the most used data types in Python because text handling is needed in almost every program.

02 String Creation

Strings can be created using single quotes, double quotes, or triple quotes.

```
s1 = "Hello"
s2 = 'Hello'
s3 = """Hello Python"""
```

Single and double quotes are commonly used for one-line strings. Triple quotes are used for multi-line strings.

```
message = """Python is simple.
Python is powerful.
Python is widely used."""
```

03 String Indexing

Indexing means accessing one character from a string. Python starts counting from 0.

```
word = "Python"
print(word[0])
print(word[3])
Output: P, h
```

For negative indexing, counting starts from the end.

```
print(word[-1])
print(word[-2])
Output: n, o
```

04 String Slicing

Slicing means taking a part of a string. The start index is included and the end index is excluded.

```
Syntax: string[start:end]
word = "Python"
print(word[0:3])
print(word[2:])
print(word[:4])
Output: Pyt / thon / Pyth
```

You can also use step values. To reverse a string:

```
print(word[0:6:2])
Output: Pto
print(word[::-1])
Output: nohtyP
```

05 String Methods

String methods are built-in functions that work on strings. They are used for cleaning, checking, changing case, finding text, and replacing text.

Method	Example → Result
lower()	"Python".lower() → "python"
upper()	"Python".upper() → "PYTHON"
strip()	" Python ".strip() → "Python"
replace()	"I like Java".replace("Java", "Python") → "I like Python"
split()	"apple,banana,mango".split(",") → [apple, banana, mango]
find()	"Python".find("th") → 2; not found → -1
count()	"banana".count("a") → 3
startswith() / endswith()	"Python".startswith("Py") → True; endswith("on") → True

06 String Formatting

String formatting means placing values inside a string in a clean way. Python has f-strings, format(), and old % formatting.

6.1 f-strings

```
name = "Aman"
age = 20
print(f"My name is {name} and I am {age} years old.")
```

6.2 format()

```
print("My name is {} and I am {} years old.".format(name, age))
```

6.3 Old % Formatting

```
print("My name is %s and I am %d years old." % (name, age))
%s = string, %d = integer
```

07 String Concatenation

Concatenation means joining strings together. The + operator is used.

```
first = "Python"
second = "Programming"
print(first + " " + second)
Output: Python Programming
```

08 Escape Characters

\n = new line \t = tab \\ = backslash \' = single quote \" = double quote

```
\n New line → print("Hello\nPython")
\t Tab → print("Name\tAge")
\\ Backslash → print("C:\\Users\\Admin")
\' Single quote → print('It\'s Python')
\" Double quote → print("He said \"Hello\"")
\b Backspace, \r Carriage Return, \f Form Feed, \v Vertical Tab, \a Alert/Bell
```

09 Raw Strings

Raw strings treat backslashes as normal characters. Write r before the string.

```
path = r"C:\new_folder\test"
print(path)
Output: C:\new_folder\test
```

10 String Multiplication

A string can be repeated using the * operator.

```
text = "Hi"
print(text * 3)
Output: HiHiHi
```

11 Common String Methods — Quick Reference

Method	Description	Example → Result
lower()	Converts all characters to lowercase	"Python".lower() → "python"
upper()	Converts all characters to uppercase	"Python".upper() → "PYTHON"
title()	Capitalizes first letter of every word	"python programming".title() → "Python Programming"
capitalize()	Capitalizes only the first letter	"python".capitalize() → "Python"
swapcase()	Swaps uppercase and lowercase	"PyThOn".swapcase() → "pYtHoN"
strip()	Removes spaces from both ends	" Python ".strip() → "Python"
lstrip()	Removes spaces from the left	" Python".lstrip() → "Python"
rstrip()	Removes spaces from the right	"Python ".rstrip() → "Python"
replace()	Replaces one substring with another	"I like Java".replace("Java", "Python") → "I like Python"
find()	Returns first index of substring	"Python".find("th") → 2
index()	Returns index; raises error if not found	"Python".index("th") → 2
count()	Counts substring occurrences	"banana".count("a") → 3
startswith()	Checks starting substring	"Python".startswith("Py") → True
endswith()	Checks ending substring	"Python".endswith("on") → True
split()	Splits string into a list	"a,b,c".split(",") → ["a", "b", "c"]
join()	Joins iterable elements into a string	"".join(["A", "B", "C"]) → "A-B-C"
isalpha()	True if all characters are alphabets	"Python".isalpha() → True
isdigit()	True if all characters are digits	"12345".isdigit() → True
isalnum()	True if letters or digits	"Python3".isalnum() → True
isspace()	True if all characters are whitespace	" ".isspace() → True
center(width)	Centers string within width	"Python".center(12) → " Python "
zfill(width)	Pads with leading zeros	"25".zfill(5) → "00025"

1-MINUTE REVISION

String = text in quotes • Index starts at 0 • Slice = [start:end] • + joins strings • * repeats strings • Methods help clean, check, change and search text